LESSON PLAN · SCRATCH (GR 3–5) BEGINNER: SCRATCH GAME MAKERS

# Lesson 10: Variables: Keeping Score

Module 4 | 45 minutes | Variables | Grades 3–5 · ages 8–10

**Objective:** Create a variable, set it at the start, change it during the game, and use it in a condition.

**Success criteria**

- I can make a variable.
- I can set it at the start and change it during the game.
- I can use a variable in a condition.

## Key vocabulary

- **variable**: A named box that stores a value.
- **value**: What is stored in the variable right now.
- **set**: Put an exact value in a variable.
- **change**: Add to (or subtract from) a variable.

## You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

## Timed lesson plan (45 min)

| Time | Stage | What happens |
|---|---|---|
| 5 min | **Warm-up (unplugged)** | Score jar quiz. |
| 5 min | **Watch** | Lesson video; pause at “Don’t forget to reset”. |
| 8 min | **Build along** | Add a score (and lives) to Catch the Apple. |
| 15 min | **Perform** | Count-the-gems puzzle, star clicker, trace the variable, matching. |
| 8 min | **Probe** | Score for Catch the Apple, Ben’s bug, explain variables. |
| 4 min | **Wrap-up** | Exit ticket. |

## Unplugged warm-up

Score jar (5 min): a jar labelled SCORE. Students answer quick questions; add a counter for each right answer. “Green flag!” → empty the jar.

## Teaching the idea

A variable is a named storage location whose value can change. Programs set (initialise) it, change it, and read it in conditions or say blocks. Initialising on green flag is essential; otherwise values

persist between runs.

**Analogy:** A variable is a labelled jar: the label is the name, the contents are the value.

**Live demo:** Build the star clicker with no reset; play twice and show the score carrying on. Add set score to 0. Then trace the “repeat 3, change by 2” example on the board with a jar and counters.

## Questions to ask

| Ask | Listen for |
|---|---|
| Where should the score be reset? | Under the green flag. |
| What is the difference between set and change? | Set chooses the exact value; change adds to it. |
| How could we use lives? | Set 3; change by −1 when missing; game over at 0. |

## Differentiation

**Support**
Pre-make the variable; students only add set and change blocks. Use the jar model.

**Stretch**
Add lives and a game-over at 0, or a high score variable that only changes when beaten.

## Common misconceptions

| Misconception | How to fix it |
|---|---|
| change score by 1 sets the score to 1 | Change ADDS 1 to whatever the score is. |
| Variables reset themselves | You must set them at the start of the game. |
| A variable name is its value | The name is the label; the value is what is inside. |

## Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

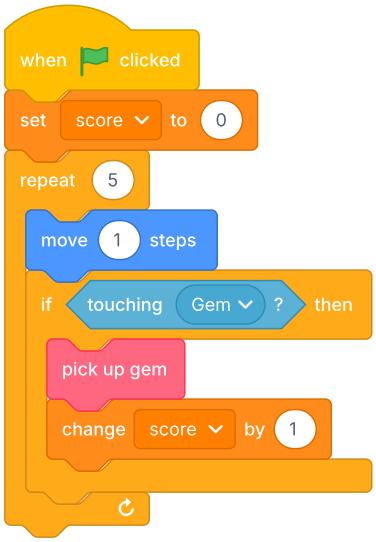
**Exit ticket**
Write the two blocks you need to keep score, and where each goes.

**Homework**
Find three things in real life that are like variables (a scoreboard, a fuel gauge…). Write what changes them.

# Answer key — Kitabash activities

## 1. Count the gems (perform · block code)

**One solution:**

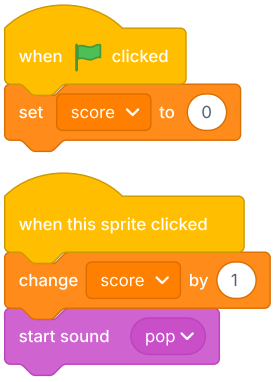



## 2. Code Lab: star clicker (perform · code lab)

**Goals checked:** Clicking adds 1 point · Green flag resets the score

**One solution:**

**Star**




## 3. Trace the variable (perform · multiple choice)

**Answer:** 5

## 4. Match each variable to what it stores (perform · matching)

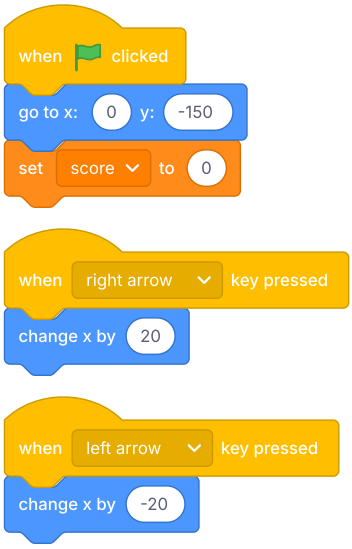
score → Points earned · lives → Chances left · timer → Seconds played · level → Which stage of the game you are on

## 5. Code Lab: score for Catch the Apple (probe · code lab)

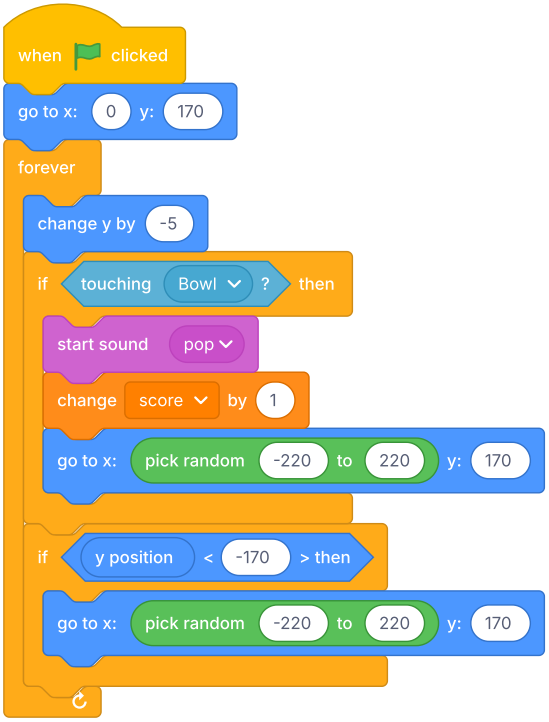
**Goals checked:** Score goes up on a catch · Score starts at 0

**One solution:**

### Bowl




### Apple




## 6. Debug it (probe · multiple choice)

**Answer:** “set score to 0” under “when green flag clicked”

**7. Explain your thinking** (probe · free response)

**Look for:** store, name, change. **Example:** A variable is like a jar with a label. The label is its name, like “sweets”, and inside it stores a number. When you eat a sweet, the number changes.

## Answer key — worksheet

1. **Trace it:** 0 · 5 · 10 · 7
2. **Set or change?:** SET · CHANGE · CHANGE
3. **Name it:** e.g. coins · timeLeft · speed
4. **Use it:** if <(lives) = (0)> then stop all (or broadcast game over).

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.